

Claims Cost Forecasting for Revenue Planning

Synthetic Data Generation Methodology

This document specifies, in full, how the raw claims and membership data for this project will be generated: the sources used to ground it, the exact distributional assumptions, the skew and cost-concentration design, the interactions and time patterns built in, and the data quality issues deliberately introduced. It is written and reviewed **before** generation, so every design choice is documented and defensible ahead of time, rather than reverse-explained after the fact.

Prepared for	Claims Cost Forecasting portfolio project
Stage	Pre-generation design lock
Output of this stage	This methodology document only — no data files yet
Next stage	Generation of claims_raw.csv and membership_raw.csv, on explicit go-ahead

1. Why Synthetic Data, and Why This Approach

Real, claim-level insurer data is not obtainable for a portfolio project: it is protected health information under HIPAA, and commercial claims datasets (HCCI, MarketScan, state APCDs) require institutional data-use agreements. The standard, industry-accepted alternative is synthetic data built to resemble real claims in structure and statistical behavior. CMS itself follows this approach with its own Data Entrepreneurs' Synthetic Public Use File (DE-SynPUF), built for exactly this purpose: letting analysts practice on realistic Medicare claims without any privacy exposure.

This project's dataset is **independently generated**, not sampled or derived from DE-SynPUF. DE-SynPUF's role is as a **structural and directional reference**: its field design (a split between payer-paid, beneficiary-responsibility, and primary-payer amounts; per-claim date spans; multiple chronic-condition flags; ICD/HCPCS coding) confirms that our schema and cost-driver design are shaped the way real claims data is shaped. We generate independently, rather than sample DE-SynPUF directly, for three reasons:

- DE-SynPUF is Medicare-only (age 65+), so its cost levels and age distribution do not represent a general commercial health-plan population, which is this project's stated scope.
- Our forecasting design requires a full billed → allowed → paid chain, claim status and denials, and a linked monthly membership/exposure table. DE-SynPUF does not provide these in a directly usable form.
- Generating independently means the true cost-generating process is known exactly, which is what allows the model's recovered coefficients and backtest accuracy to be checked against ground truth later in the project — impossible with any dataset whose true process is unknown.

What makes this defensible, not arbitrary: every distribution, interaction and skew pattern below is chosen to match well-documented, publicly known facts about how U.S. healthcare claims behave (inpatient claims are rare but dominate total dollars; costs are right-skewed; larger claims arrive later; older and chronically ill members cost more), rather than picked without a stated reason. Section 4 lists the specific numeric assumptions so they can be checked line by line.

2. Source Lineage

The final column list is a merge of three inputs, so every field in the generated data can be traced to a reason for existing.

Source	What it contributed
Original claims_raw.csv (this project's starting point)	Core per-claim features: patient age, prior claim history, procedure complexity, length of stay, chronic flag,
CMS DE-SynPUF (structural reference)	The billed/allowed/paid-style financial split; per-claim date spans; multiple chronic-condition flags instead of
Gaps identified during scoping (this project's design review)	Service/submission/paid dates, claim status and denials, adjustments and restatements, a separate memb

3. Final Raw Schema (Pre-SQL, Ungrouped)

Both files are generated flat, exactly as they would land in a staging table from two separate source systems (claims processing and enrollment). No fact/dimension normalization happens at this stage — that is deliberately left for the SQL layer.

3.1 claims_raw.csv

Column	Description
claim_id	Claim identifier. Not guaranteed unique (duplicates injected, Section 5).
member_id	Links to membership_raw.csv. A small share will not match (orphan records, Section 5).
provider_id / provider_npi	Provider identifier; drives provider-level cost variation.
service_date	Date care was provided (the 'incurred' date).
claim_submission_date	Date the claim was submitted, after service_date by a claim-type-dependent lag.
paid_date	Date paid; null for pending/denied claims.
diagnosis_code_1 / diagnosis_code_2	ICD-10-style codes; diagnosis_code_2 mostly null.
procedure_code	CPT/HCPCS-style code.
revenue_code	Inpatient-only field; null elsewhere (realistic sparsity by claim type).
claim_type	Inpatient, Outpatient, Emergency, Pharmacy, Preventive.
place_of_service	Setting descriptor, correlated with claim_type.
length_of_stay	Inpatient/emergency only; 0 or null for others.
billed_amount / allowed_amount / paid_amount	Full financial chain; paid_amount is the model target.
patient_responsibility_amount	Deductible + coinsurance portion.
claim_status	Raw, inconsistent text: 'Paid', 'paid', 'PD', 'Denied', 'DENIED', 'Pending'.
denial_reason_code	Populated only when claim_status indicates denial.
provider_type / provider_specialty / network_id	Provider descriptors.
patient_age / patient_sex / region	Denormalized patient fields, as real claim extracts often carry.
chronic_flag_diabetes / chf / cancer / covid_severe	Five separate binary flags (DE-SynPUF-style).
adjustment_flag / original_claim_id	Marks corrections; original_claim_id links back to the claim being adjusted.
record_source / load_timestamp	Source-system metadata; deliberately inconsistent formatting.

3.2 membership_raw.csv

Column	Description
member_id	Member identifier.
year_month	One row per member per enrolled month.
plan_type	e.g. HMO, PPO, HDHP.
region	Inconsistently spelled across rows on purpose (Section 5).
enrollment_status	Active, Termined, etc.
birth_date	Used to derive age at time of each claim.
sex	Member sex.

3.3 Sizing

File	Rows	Represents
claims_raw.csv	1,000,000 claims across ~3 years	Raw claims-system extract
membership_raw.csv	~50,000 members × up to 36 months ≈ 1.6 million	Raw membership-system extract

1,000,000 claims over 3 years against ~50,000 members works out to roughly 6–7 claims per member per year, in line with typical mixed-population utilization, with heavy skew toward a small number of high utilizers and a large share of members with few or no claims in a given year.

4. Cost Skew and Concentration Design

The single most important property of claims data — the one the entire log-transform and GLM discussion in this project rests on — is that cost is heavily right-skewed and dollar-concentrated in a small share of claims. This is designed in explicitly, not left to chance.

4.1 Base severity distribution, by claim type

Paid amount is drawn from a lognormal distribution per claim type (before interactions/adjustments below are applied). Lognormal parameters (μ , σ of the underlying normal) are chosen so that the resulting medians and right-tail behavior match the well-documented real-world pattern that inpatient claims are the minority of volume but the majority of dollars.

Claim type	Target volume share	Lognormal median	Approx. mean	Design intent
Inpatient	~8-10%	~\$9,000	~\$14,000+	Heaviest right tail; drives most dollar concentration
Emergency	~10%	~\$1,800	~\$2,600	Moderate skew
Outpatient	~38-40%	~\$650	~\$950	Moderate skew
Pharmacy	~25%	~\$90	~\$140	Low skew, high volume
Preventive	~14-16%	~\$180	~\$220	Low skew, capped upside

An additional **catastrophic tail** is layered on top: roughly 0.3–0.5% of Inpatient and Emergency claims are redrawn from a Pareto-style heavy tail reaching \$150,000–\$400,000+, representing trauma, transplant, or extended-ICU cases. This is what will make the top ~1% of claims responsible for roughly 20–25% of total dollars, mirroring the concentration already observed when the original dataset was profiled.

4.2 Interactions and multipliers applied on top of the base draw

- **Chronic condition load:** each active chronic flag (diabetes, CHF, cancer, COPD, ESRD) applies a multiplicative cost increase (roughly $\times 1.1$ to $\times 1.4$ per flag, compounding, with cancer and ESRD weighted highest), consistent with real chronic-disease cost burden.
- **Chronic $\times$ procedure complexity interaction:** the effect of procedure complexity on cost is amplified for chronically ill patients, not merely additive — this is the specific non-linear interaction the XGBoost-vs-linear-model benchmark is designed to detect.
- **Length of stay:** for Inpatient/Emergency claims, cost scales super-linearly with length of stay (each additional day costs more than the last, reflecting escalating acuity), rather than a flat per-day rate.
- **Age:** a U-shaped cost curve — elevated cost for very young (under 2) and older (65+) age bands relative to mid-life, rather than a simple linear age effect.
- **Network status:** out-of-network claims cost 20–40% more on billed/allowed amounts, reflecting the absence of a negotiated rate.
- **Regional cost index:** a fixed per-region multiplier (roughly $\pm 15\%$), reflecting real regional cost-of-care variation.
- **Provider-level effect:** individual providers (not just provider_type categories) carry their own persistent cost tendency, so two providers of the same type are not interchangeable — this supports later provider-level analysis.

4.3 Time patterns

- **Trend:** underlying cost levels increase ~5–6% year-over-year across the 3-year window, consistent with typical medical cost trend.
- **Seasonality:** a winter uplift (roughly November–February) in Inpatient and Emergency claim volume and severity, reflecting respiratory/flu-season patterns.
- **Claim lag:** submission_date trails service_date by a claim-type-dependent number of days (Pharmacy fastest, Inpatient slowest), and paid_date trails submission_date similarly — this is what generates a realistic claim-lag triangle for the completion-factor/IBNR analysis.

5. Deliberate Data Quality Issues

Real claims extracts are never clean. The issues below are injected at stated, documented rates so that the SQL cleaning and reconciliation layer has genuine, quantifiable problems to find and fix — and so that its output can be checked ("did we recover 100% of true dollars after removing exact duplicates?") against a known injection rate.

Issue	Approx. rate	Where
Exact duplicate claim_id rows	~1.5%	claims_raw
Missing procedure_complexity-relevant fields (proc./diag. code null)	~1–5%	claims_raw
Impossible / negative patient_age	~0.3%	claims_raw
paid_amount > billed_amount (processing error)	~0.5%	claims_raw
Inconsistent claim_status casing/text ('Paid'/'paid'/'PD')	throughout	claims_raw
Orphan member_id (no matching membership row)	~2–3%	claims_raw
Malformed / inconsistent load_timestamp	~2%	claims_raw
Adjustment/reversal records referencing an original claim	~1.5%	claims_raw
Missing enrollment month (coverage gap or load miss — ambiguous design)	3–4%	membership_raw
Duplicate member-month rows	~1%	membership_raw
Inconsistent region spelling (e.g. 'NE'/'Northeast'/'North East')	throughout	both files

All injection rates above are logged internally at generation time (not published in the raw files themselves), so that after the SQL cleaning layer runs, its results can be checked against the true injected counts as a validation step in its own right.

6. Why This Design Is Defensible

- Every distributional and interaction choice above is tied to a documented, real-world claims pattern, not picked arbitrarily — so each one has a one-line justification that survives a follow-up question in an interview.
- The true cost-generating process is known exactly, which lets every later modeling claim ("the GLM recovered the chronic × complexity interaction", "the log-linear model missed the LOS non-linearity that boosting caught") be checked against ground truth, not asserted.
- The schema and financial structure are aligned to CMS DE-SynPUF's real design, so the project's realism claim rests on a named, public reference rather than "looks reasonable to me."
- Data quality issues are injected at known, stated rates, which turns the SQL cleaning/reconciliation layer into something independently verifiable rather than cosmetic.

The honest limitation, to be stated plainly in the project's final report: this is a designed synthetic dataset, calibrated to known patterns in the literature and to CMS's own public synthetic-data structure, not sampled from or validated against a real commercial insurer's book of business. It is built to teach and demonstrate the correct method, not to produce numbers that represent any real population.

7. Status

This document is the locked design. No data has been generated. Generation of `claims_raw.csv` and `membership_raw.csv` proceeds only on explicit confirmation to generate, using exactly the parameters specified above.